1. $\rho_k(k)dk = 2\times\frac{1}{2}\times\frac{dk}{\frac{\pi}{a}} = \frac{a}{\pi}dk.$

因为 $k^2 = \frac{2mE}{\hbar^2}$

则 $k = \frac{1}{\hbar}\sqrt{2mE}$, $dk = \frac{1}{\hbar}\sqrt{\frac{m}{2E}}\,dE$

则 $\rho_E(E)\,dE = \frac{a}{\pi}\cdot\frac{1}{\hbar}\sqrt{\frac{m}{2E}}\cdot dE$

即 $\rho(E) = \frac{a}{\pi\hbar}\sqrt{\frac{m}{2E}}.$

2. (a) $R = \frac{U}{I} = \frac{10V}{100mA} = 100\Omega.$

(b) $R = \rho\cdot\frac{L}{S} = \frac{L}{\sigma S}$

则 $\sigma = \frac{L}{SR} = \frac{1\times10^{-3}cm}{1\times10^{-3}cm^2\times100\Omega} = 1\ S\cdot m^{-1} = 0.01\ S\cdot cm^{-1}$

(c) $\sigma = ne\mu_n.$

$T=300K$ 时, $\mu_n \approx 1350\ cm^2/(V\cdot s)$

$n = \frac{\sigma}{e\mu_n}$

$= \frac{0.01}{1.6\times10^{-19}\times1350}$

$= 4.63\times10^{13}\ cm^{-3}$

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